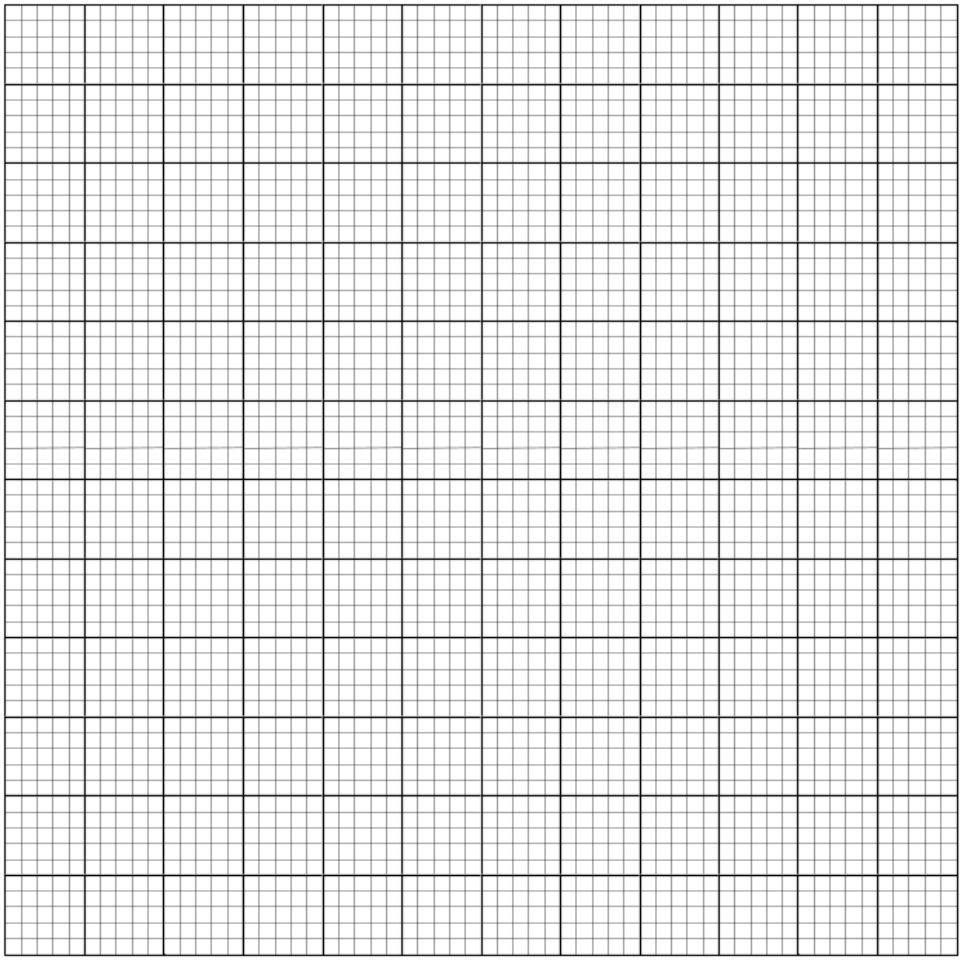
Science Homework: **Motion Graphs** **Due on:**

Name: _____ Mark: _____/22

1) A car is monitored during a journey through town, the data collected was:

Time (hours)	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	6.0
Distance (km)	0	45	90	135	135	155	175	120	75	75	60	30

a) Plot the distance/time graph below.



(5 marks)

b) Calculate the velocity:

i) In the first 1.5 hours _____

_____(_____)

(2 marks)

ii) Between 3.0 and 4.5 hours _____

_____(_____)

(2 marks)

iii) Averaged for the whole journey _____

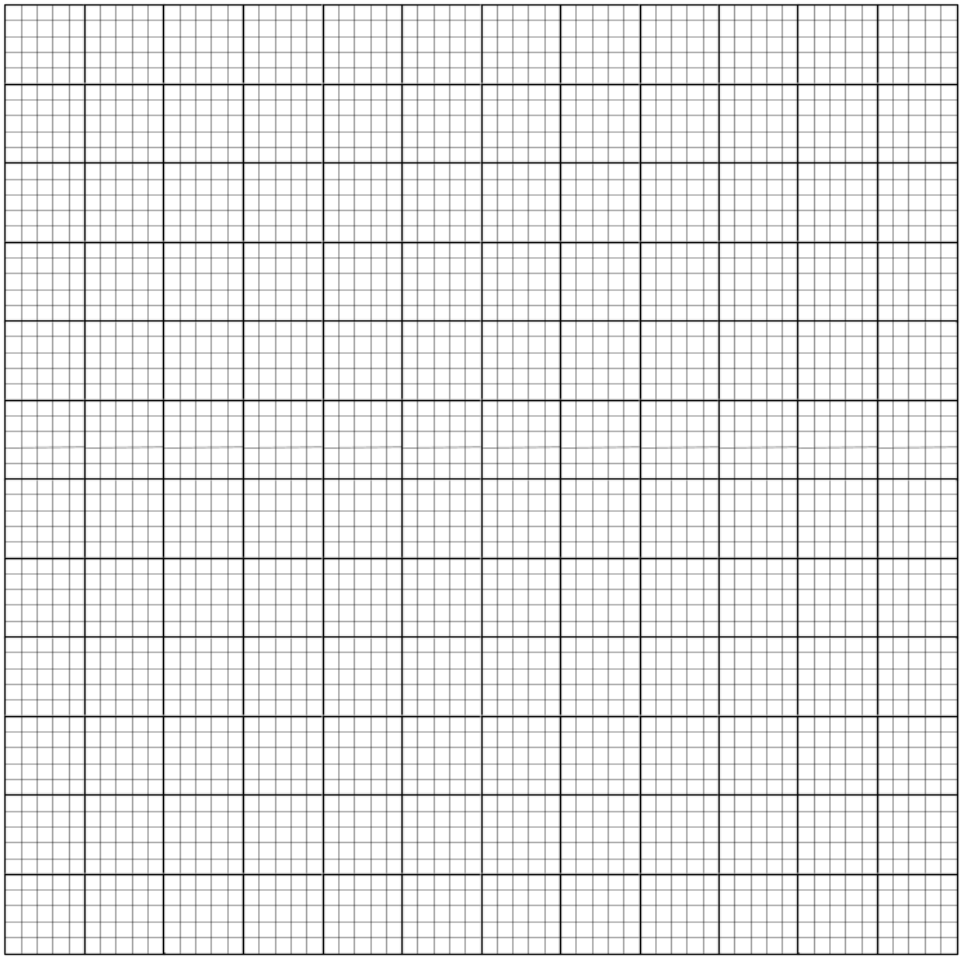
_____(_____)

(2 marks)

2) The velocity data for an Olympic sprinter is shown below:

Time (s)	0	1	2	3	4	5	6	7	8	9	10	11
velocity (ms^{-1})	0.0	3.4	7.8	10.2	11.2	11.0	11.0	10.8	10.6	10.5	10.4	6.4

a) Plot the velocity/time graph below.



(5 marks)

b) Calculate the acceleration:

i) In the first 3.0 seconds _____

_____(_____)

(2 marks)

ii) In the last second _____

_____(_____)

(2 marks)

c) Describe the motion of the sprinter

_____

_____

_____

(2 marks)